Asadullah Bhuiyan

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Available for full-time roles starting July 2027.

Education

Cornell University Aug 2021 – Jun 2027 (expected)
Ph.D. in Theoretical Physics. Advisor: Chao-Ming Jian.
Thesis: *Random Processes in Topological Quantum Dynamics and Machine Learning*.

University of Alberta 2017 – 2021
B.Sc. in Engineering Physics, with distinction. GPA: 3.9/4.0

Research & Technical Experience

Doctoral Research, Cornell University May 2022 – Present

Analytic theory of neural-network learning under input corruption Paper | Code

- Initiated an independent, student-led neural-network project, set the research direction, and recruited three collaborators.
- Built PyTorch pipelines to train and analyze ensembles of finite-width MLPs under varying levels of training-input corruption.
- Derived a leading-order, interpretable decision rule universal across MLP architectures in the high-noise, infinite-width limit using symmetry, linear response, and neural tangent kernel (NTK) theory.

Topology and entropy in stochastic quantum dynamics Paper | Code

- Established a Lie-algebraic classification that organizes Gaussian dynamical system ensembles into ten symmetry classes; proved, for each class, whether symmetry-respecting Gaussian measurement-operator ensembles can exist.
- Designed and analyzed a stochastic, nonlinear quantum control protocol for preparing any fermionic Gaussian topological phase in arbitrary spatial dimensions.
- Formulated an effective description of the dynamics and analytically derived relaxation rates via covariance matrix methods.
- Simulated the protocol's stochastic real-time covariance evolution and topological phase transitions using Python/PyTorch.
- Proved an exponential bound on mean second Rényi entropy in Gaussian circuits with Haar-random passive unitaries and single-mode measurements; constructed a replica spin-chain theory for $1+1$D entropy dynamics.

Undergraduate Research, University of Alberta May 2018 – Aug 2021

- Developed an iterative MATLAB solver for a system of five Green's-function convolutions on a 3D mesh in the 2D Hubbard-Holstein model; coauthored four peer-reviewed publications across five projects.

Product Engineering Intern, Microchip Technology Inc. Sep 2019 – Dec 2019

- Designed hardware reliability tests under extreme electrical and thermal stress, built an automated Python pipeline for test execution and statistical data analysis, and reported findings to engineering leadership.

Technical Skills

Programming & scientific computing: Python (PyTorch, NumPy, SciPy, pandas, Matplotlib), MATLAB, Mathematica, Linux, Git.
Quantitative methods & AI workflows: Stochastic processes, covariance-matrix methods, replica & random matrix theory, numerical linear algebra; agentic coding (OpenAI Codex, Claude Code).

Selected Publications (2 of 6)

J. Tahmassebpur, **A. Bhuiyan**, H. Kim, and O. Lesser. Neural networks become centroid classifiers when trained on heavily corrupted inputs. *Physical Review Research* **8**, 043088 (2026).
A. Bhuiyan, H. Pan, and C.-M. Jian. Free-Fermion Dynamics with Measurements: Topological Classification and Adaptive Preparation of Topological States. *Physical Review Research* **8**, 023147 (2026).

Honors, Talks, and Training

Quantum Nano Seminar, Dartmouth College (2025). Invited research seminar.
Fellowships: Cornell Graduate Fellowship (2025); NSERC PGS-D Three-Year Doctoral Scholarship (2022–2025).
Selective Workshops: Institute for Advanced Study (2023); Boulder School for Condensed Matter Theory (2022).

Teaching and Mentoring

Teaching, Cornell: Seminar instructor, AEP 4230 (2026) and PHYS 1201 (2021); seminar/lab instructor, PHYS 2214 (2022). Developed original coursework connecting statistical physics and neural networks.
Mentoring: Student advisor for undergraduate and first-year graduate physics students (2023–2025); mentored a high-school student's award-winning Qiskit entanglement-dynamics project (2024–2025).